

Text Document

Generated on: 7/15/2026

```
import pandas as pd
```

```
import joblib
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
# Load dataset
```

```
data = pd.read_csv("../Dataset/flood_dataset_classification.csv")
```

```
print("Dataset Loaded")
```

```
print(data.head())
```

```
# Separate features and target
```

```
X = data.drop("occured", axis=1)
```

```
y = data["occured"]
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
```

```
    y,
```

```
    test_size=0.2,
```

```
    random_state=42
```

```
)
```

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```
# Create Machine Learning model
```

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)
```

```
# Train model
```

```
model.fit(X_train, y_train)
```

```
# Test model
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(  
    y_test,  
    y_pred  
)
```

```
print("Model Accuracy:", accuracy)
```

```
# Save trained model
```

```
joblib.dump(model, "flood_model.pkl")
```

```
print("flood_model.pkl created successfully")
```